

Research Notes on Visual question answering

Visual question answering

>> Section 1

Mathematics An open-source, math-aware, question answering system called MathQA, based on Ask Platypus and Wikidata, was published in 2018. MathQA takes an English or Hindi natural language question as input and returns a mathematical formula retrieved from Wikidata as a succinct answer, translated into a computable form that allows the user to insert values for the variables. The system retrieves names and values of variables and common constants from Wikidata if those are available. It is claimed that the system outperforms a commercial computational mathematical knowledge engine on a test set. MathQA is hosted by Wikimedia at <https://mathqa.wmflabs.org/>. In 2022, it was extended to answer 15 math question types. MathQA methods need to combine natural and formula language. One possible approach is to perform supervised annotation via Entity Linking. The "ARQMath Task" at CLEF 2020 was launched to address the problem of linking newly posted questions from the platform Math Stack Exchange to existing ones that were already answered by the community. Providing hyperlinks to already answered, semantically related questions helps users to get answers earlier but is a challenging problem because semantic relatedness is not trivial. The lab was motivated by the fact that 20% of mathematical queries in general-purpose search engines are expressed as well-formed questions. The challenge contained two separate sub-tasks. Task 1: "Answer retrieval" matching old post answers to newly posed questions, and Task 2: "Formula retrieval" matching old post formulae to new questions. Starting with the domain of mathematics, which involves formula language, the goal is to later extend the task to other domains (e.g., STEM disciplines, such as chemistry, biology, etc.), which employ other types of special notation (e.g., chemical formulae). The inverse of mathematical question answering—mathematical question generation—has also been researched. The PhysWikiQuiz physics question generation and test engine retrieves mathematical formulae from Wikidata together with semantic information about their constituting identifiers (names and values of variables). The formulae are then rearranged to generate a set of formula variants. Subsequently, the variables are substituted with random values to generate a large number of different questions suitable for individual student tests.

>> Section 2

Answering questions related to an article in order to evaluate reading comprehension is one of the simpler form of question answering, since a given article is relatively short compared to the domains of other types of question-answering problems. An example of such a

question is "What did Albert Einstein win the Nobel Prize for?" after an article about this subject is given to the system. Closed-book question answering is when a system has memorized some facts during training and can answer questions without explicitly being given a context. This is similar to humans taking closed-book exams. Closed-domain question answering deals with questions under a specific domain (for example, medicine or automotive maintenance) and can exploit domain-specific knowledge frequently formalized in ontologies. Alternatively, "closed-domain" might refer to a situation where only a limited type of questions are accepted, such as questions asking for descriptive rather than procedural information. Question answering systems in the context of machine reading applications have also been constructed in the medical domain, for instance related to Alzheimer's disease. Open-domain question answering deals with questions about nearly anything and can only rely on general ontologies and world knowledge. Systems designed for open-domain question answering usually have much more data available from which to extract the answer. An example of an open-domain question is "What did Albert Einstein win the Nobel Prize for?" while no article about this subject is given to the system. Another way to categorize question-answering systems is by the technical approach used. There are a number of different types of QA systems, including:

>> Section 3

interactivity—clarification of questions or answers answer reuse or caching semantic parsing answer presentation knowledge representation and semantic entailment social media analysis with question answering systems sentiment analysis utilization of thematic roles Image captioning for visual question answering Embodied question answering In 2011, Watson, a question answering computer system developed by IBM, competed in two exhibition matches of Jeopardy! against Brad Rutter and Ken Jennings, winning by a significant margin. Facebook Research made their DrQA system available under an open source license. This system uses Wikipedia as knowledge source. The open source framework Haystack by deepset combines open-domain question answering with generative question answering and supports the domain adaptation of the underlying language models for industry use cases. Large Language Models (LLMs)[36] like GPT-4[37], Gemini[38] are examples of successful QA systems that are enabling more sophisticated understanding and generation of text. When coupled with Multimodal[39] QA Systems, which can process and understand information from various modalities like text, images, and audio, LLMs significantly improve the capabilities of QA systems.

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>> Section 4

Methods Question answering is dependent on a good search corpus; without documents containing the answer, there is little any question answering system can do. Larger collections generally mean better question answering performance, unless the question domain is orthogonal to the collection. Data redundancy in massive collections, such as the web, means that nuggets of information are likely to be phrased in many different ways in differing contexts and documents, leading to two benefits:

>> Section 5

Architecture As of 2001, question-answering systems typically included a question classifier module that determined the type of question and the type of answer. Different types of question-answering systems employ different architectures. For example, modern open-domain question answering systems may use a retriever-reader architecture. The retriever is aimed at retrieving relevant documents related to a given question, while the reader is used to infer the answer from the retrieved documents. Systems such as GPT-3, T5, and BART use an end-to-end architecture in which a transformer-based architecture stores large-scale textual data in the underlying parameters. Such models can answer questions without accessing any external knowledge sources.

>> Section 6

If the right information appears in many forms, the question answering system needs to perform fewer complex NLP techniques to understand the text. Correct answers can be filtered from false positives because the system can rely on versions of the correct answer appearing more times in the corpus than incorrect ones. Some question answering systems rely heavily on automated reasoning.